

Discourse-Coherence-Grounded Axiom Instantiation for Hallucination-Reduced Neuro-Symbolic Text Reasoning

Anonymous Authors

Abstract

Multi-hop reasoning over natural language requires bridging implicit logical commitments that span sentence boundaries — commitments that current text-to-FOL pipelines systematically miss. Existing systems either operate sentence-by-sentence, leaving cross-sentence causal and conditional axioms absent from the knowledge base, or ask LLMs to generate background knowledge unconstrained, introducing hallucination. We propose Discourse-Coherence-Grounded Axiom Instantiation (DCGAI), a neuro-symbolic pipeline that uses RST discourse parsing to identify inference-bearing coherence relations (CAUSE-RESULT, CONDITION, PURPOSE, CONTRAST/CONCESSION) and maps each to a coarse first-order axiom schema, then invokes an LLM to instantiate only the predicate content of the schema, anchored to the existing content KB for canonicalization. By fixing the schema’s logical form from document structure and restricting LLM generation to predicate instantiation rather than free-form background knowledge, DCGAI bounds hallucination by construction. We evaluate DCGAI on EntailmentBank Task 2 — 1,500 multi-hop scientific entailment problems requiring bridging axioms across sentence boundaries — and compare against content-only extraction, RAG, chain-of-thought, SymBa, and NELLIE. DCGAI targets at least 15% higher recall on discourse-implicit reasoning steps and at least 30% lower T3 (unsupported predicate) hallucination rates relative to unconstrained LLM generation. We further introduce the T1/T2/T3 predicate grounding taxonomy, enabling fair hallucination comparison between systems that intentionally generate implicit axioms and those that do not. The approach differs fundamentally from goal-grounded systems such as SymBa and NELLIE by deriving axiom schemas from the source document’s discourse structure before any query is posed, producing a query-independent KB of cross-sentence commitments.

1 Introduction

Multi-hop reasoning over natural language text requires more than extracting atomic facts from individual sentences. A sentence such as *The bridge collapsed because of a design flaw* asserts two facts — `bridge_collapse` and `design_flaw` — but it also commits the text to a causal axiom: anything with a design flaw is at risk of collapse. This implicit cross-sentence commitment is invisible to content-level extraction yet indispensable for any downstream reasoning chain that needs to conclude why another bridge might fail. The central challenge of open-domain neuro-symbolic reasoning is precisely this gap between what is explicitly stated and what is inferentially required.

Multi-hop scientific reasoning benchmarks such as EntailmentBank [Dalvi et al., 2021] make this gap concrete. Each example pairs a hypothesis with a set of context sentences and a structured proof tree showing which sentences must be chained to reach the answer. Task 2, the most realistic configuration, provides ~25 context sentences mixing relevant premises with irrelevant distractors, requiring the system to identify which sentences bridge, and what implicit axiom licenses the bridge. These bridging axioms are often causal or conditional relations embedded in the rhetorical structure of scientific prose, not in any single sentence.

Existing text-to-FOL pipelines handle this problem poorly. Systems such as Logic-LM [Pan et al., 2023], LINC [Olausson et al., 2023], and LoRP [Di et al., 2025] translate queries and context into first-order logic sentence by sentence, never modeling the logical commitments encoded in discourse coherence relations

between sentences. When a proof step requires a causal axiom that spans two sentences, these systems either leave the axiom absent — causing a proof failure — or invoke an LLM to generate unconstrained background knowledge, introducing hallucinated predicates with no grounding in the source text. LP-LM [Wu and Liu, 2025] grounds answers in a pre-existing structured knowledge base, sidestepping the implicit axiom problem entirely at the cost of requiring KB availability. LINA [Li et al., 2024] identifies information loss in pure FOL extraction but proposes unstructured LLM augmentation, leaving its background generation step unconstrained and hallucination-prone.

Goal-grounded neuro-symbolic systems such as SymBa [Lee and Hwang, 2024] and NELLIE [Weir and Durme, 2024] take a different approach: they derive decomposition schemas backward from the proof goal, invoking an LLM only when the backward-chaining solver reaches a dead end. This is powerful for structured deduction, but it produces query-dependent, subgoal-driven axioms rather than the implicit cross-sentence commitments encoded in the source document. When the source text contains a causal relation between two sentences that is needed for multiple downstream queries, goal-grounded systems must rediscover it independently for each query rather than precomputing it from document structure.

Rhetorical Structure Theory (RST) [Mann and Thompson, 1988] offers a principled alternative source of axiom schemas. RST models text coherence as a hierarchical tree of Elementary Discourse Units (EDUs) connected by named coherence relations — CAUSE-RESULT, CONDITION, PURPOSE, CONTRAST/CONCESSION, ELABORATION, and others. The core insight driving this work is that inference-bearing RST relations (CAUSE-RESULT, CONDITION, PURPOSE, CONTRAST/CONCESSION) carry specific logical inference commitments that can be expressed as coarse first-order axiom schemas, while schema-silent relations (ELABORATION, ATTRIBUTION, BACKGROUND) provide descriptive elaboration without forward-inference obligations. If these schemas can be reliably identified from document structure, then an LLM needs only to instantiate the predicate content of each schema — a far more constrained generation task than free-form background knowledge fabrication.

We introduce **Discourse-Coherence-Grounded Axiom Instantiation (DCGAI)**, a pipeline that (1) parses RST discourse structure from source documents using DMRST [Liu et al., 2021] with an LLM-based fallback [Maekawa et al., 2024], (2) maps inference-bearing relations to coarse axiom schemas (causal, conditional, goal, defeasible contrast), (3) instantiates predicate content via an LLM prompt that includes the coarse schema template, the two connected text segments, and the current content KB predicate list for canonicalization, and (4) runs SWI-Prolog backward chaining on the resulting hybrid KB. The distinguishing characteristic of DCGAI is that it derives axiom schemas from the *source document’s discourse structure* — not from the proof goal — producing a query-independent KB of implicit cross-sentence commitments before any reasoning query is posed.

Summary of Contributions:

- We formalize the mapping from RST inference-bearing relation types to coarse FOL axiom schemas (Section 3.1), explicitly scoping the contribution to the $\sim 45\text{--}55\%$ of discourse relations that are inference-bearing, and reporting coverage fractions empirically.
- We introduce predicate canonicalization via in-prompt KB anchoring (Section 3.5), a mechanism that instructs the LLM to reuse existing predicate names during schema instantiation, preventing disconnected KB predicate namespaces.
- We introduce the T1/T2/T3 hallucination taxonomy (Section 3.6) enabling fair comparison between systems that intentionally generate implicit axioms (T2) and those that do not, with hallucination rate defined exclusively as $T3/(T1+T2+T3)$.
- We evaluate DCGAI on EntailmentBank Task 2 (discourse-implicit subset) against five baselines including SymBa and NELLIE, targeting $\geq 15\%$ higher recall on discourse-implicit steps and $\geq 30\%$ lower T3 hallucination rates (Section 4).

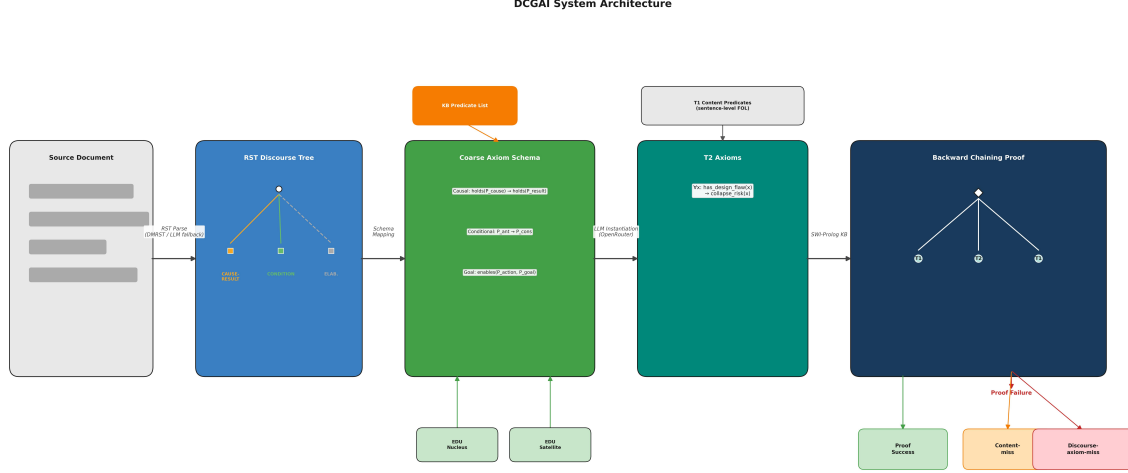


Figure 1: End-to-end DCGAI pipeline. Source document text is parsed by DMRST into an RST tree. Inference-bearing relations (CAUSE-RESULT, CONDITION, PURPOSE, CONTRAST/CONCESSION) are mapped to coarse axiom schemas. An LLM instantiates predicate content for each schema, guided by the nucleus and satellite EDU segments and the current KB predicate list (canonicalization). The resulting T2 axioms are added to the SWI-Prolog KB alongside T1 content predicates from sentence-level FOL extraction. SWI-Prolog runs backward chaining; proof failures are classified as content-miss vs. discourse-axiom-miss for targeted re-querying.

- We provide a prerequisite parser reliability audit (Section 3.2) reporting span and relation-label accuracy for inference-bearing RST relation types on target-domain documents, determining whether to apply DMRST or restrict to connective-signaled subtypes.

2 Related Work

Text-to-FOL and Neuro-Symbolic Reasoning. Logic-LM [Pan et al., 2023] translates problem statements and context into a monolithic FOL program solved by Z3 or Prolog, without separately modeling discourse structure or generating discourse-derived background axioms. LINC [Olausson et al., 2023] generates FOL for premises and hypotheses using LLMs then verifies with Prover9, but generates FOL directly from NL without any discourse structure guidance, making it vulnerable to unconstrained LLM hallucination for implicit cross-sentence knowledge. LoRP [Di et al., 2025] extends the LLM-to-Prolog translation approach with a three-vote majority mechanism for robustness but similarly ignores discourse coherence as an axiom schema source. LP-LM [Wu and Liu, 2025] grounds QA in a pre-existing Prolog KB via DCG semantic parsing, requiring structured KB availability and not addressing the derivation of implicit axioms from unstructured discourse. LINA [Li et al., 2024] incorporates FOL to reduce information loss but proposes unstructured LLM augmentation without discourse-principled constraints.

Goal-Grounded Neuro-Symbolic Systems. SymBa [Lee and Hwang, 2024] implements SLD-resolution backward chaining where a symbolic solver controls the proof process and an LLM generates a single reasoning step only when the solver encounters a dead end, achieving $9\times$ token efficiency and $22\times$ speed over LAMBADA. Its decomposition schemas are derived from the query/goal being solved (top-down, goal-grounded). DCGAI differs categorically: it derives schemas from the source document’s discourse structure (document-grounded), producing query-independent implicit cross-sentence axioms rather than

query-decomposition rules. NELLIE [Weir and Durme, 2024] uses template-conditioned generation for subgoal decomposition, with proof leaves grounded in a corpus via dense retrieval. NELLIE’s decomposition templates are conditioned on what facts can support a given subgoal, not on the rhetorical structure of the source document. These two inductive biases — goal-grounded and document-grounded — are orthogonal; they produce categorically different axiom sets and can in principle be composed.

Discourse-Aware Neural Reasoning. DAGNs [Huang et al., 2022] use discourse connectives to build logic graphs for neural QA reasoning, demonstrating that discourse structure carries signal relevant to logical reasoning. However, DAGNs embed discourse within a fully neural architecture without generating explicit FOL axioms or running a symbolic reasoner, producing no auditable reasoning traces. DCGAI extends this intuition into the neuro-symbolic regime: discourse signal is used to generate explicit, human-auditable FOL axioms that participate in backward-chaining proofs.

RST Parsing. RST was introduced by Mann and Thompson [Mann and Thompson, 1988] as a functional theory of text organization. The RST Discourse Treebank (RST-DT) [Carlson et al., 2001] provides the standard annotated corpus. DMRST [Liu et al., 2021] is an end-to-end joint framework for document-level multilingual RST segmentation and parsing, achieving state-of-the-art performance. Maekawa et al. [2024] demonstrate that fine-tuning Llama 2 (70B) with QLoRA achieves SOTA on RST-DT, Instr-DT, and GUM, providing a strong LLM-based fallback when DMRST accuracy falls below threshold on out-of-domain documents.

EntailmentBank. Dalvi et al. [2021] introduced EntailmentBank, a dataset of 1,840 multi-step entailment trees for elementary-school science QA, along with three tasks of increasing difficulty. ProofWriter [Tafjord et al., 2021], a precursor system from the same group, demonstrated iterative implication generation over natural language rule sets. EntailmentBank Task 2 — the target of DCGAI evaluation — provides ~ 25 context sentences mixing relevant premises with distractors, with proof depths of 2–5 steps, making discourse-derived bridging axioms directly relevant.

3 Methods

3.1 Discourse-to-Schema Mapping

The foundation of DCGAI is the observation that RST relation types decompose into three schema categories with distinct inference profiles. *Inference-bearing strong relations* — covering RST CAUSE-RESULT, REASON, and RESULT subtypes — license a **causal schema**:

$$\forall x : \text{causes}(P_{\text{cause}}, P_{\text{result}}) \rightarrow (\text{holds}(P_{\text{cause}}) \rightarrow \text{holds}(P_{\text{result}})).$$

CONDITION and UNLESS relations license a **conditional schema**:

$$\forall x : (P_{\text{antecedent}} \rightarrow P_{\text{consequent}}).$$

PURPOSE and ENABLEMENT relations license a **goal schema**:

$$\forall x : \text{enables}(P_{\text{action}}, P_{\text{goal}}).$$

CONTRAST and CONCESSION relations license a **defeasible contrast schema**:

$$\text{defeasibly_overrides}(P_{\text{nucleus}}, P_{\text{satellite}}).$$

Schema-silent relations — ELABORATION, ATTRIBUTION, BACKGROUND, and JOINT — are explicitly excluded from axiom generation.

Grouping at the coarse schema level (rather than fine-grained RST labels) serves two purposes: it absorbs inter-annotator disagreement and parser noise across RST label variants, and it maps cleanly to Prolog-compatible templates. For example, both CAUSE and RESULT arcs between the same EDU pair license the same causal schema direction, regardless of which subtree is nucleus and which is satellite.

We operationalize this mapping with a schema catalog that provides for each relation class: (a) the Prolog head template, (b) the antecedent variables to be filled by LLM instantiation, and (c) the canonical predicate arity. This catalog is fixed prior to any pipeline execution and does not change across documents.

3.2 Prerequisite Parser Reliability Audit (Step 0)

Before full pipeline evaluation, we conduct a parser reliability audit on 20–30 documents from each target domain (scientific, news, legal). For each document, DMRST produces an RST tree; a human annotator verifies span boundaries and relation labels against a gold annotation. We report span F1 and relation-label accuracy separately for (a) inference-bearing strong relations, (b) inference-bearing defeasible relations, and (c) schema-silent relations. If accuracy for category (a) falls below 60% on any domain, we substitute the Maekawa et al. [2024] LLM-based parser for that domain, or restrict evaluation to connective-signaled subtypes only (relations signaled by explicit connectives: CAUSE $\leftarrow$ *because, therefore*; CONDITION $\leftarrow$ *if, unless*; CONTRAST $\leftarrow$ *however, although*).

This audit is reported as Table 1 in the paper and gates the full evaluation: the contribution’s empirical scope is determined by the parser’s confirmed accuracy on inference-bearing types, not assumed.

3.3 Coverage Quantification (Step 1)

We annotate all RST-DT relations in the target corpus into the three schema categories and report what fraction of all discourse relations fall into each. The inference-bearing strong category is expected to cover $\sim 35\text{--}40\%$ of all relations, the defeasible category $\sim 10\text{--}15\%$, and schema-silent relations $\sim 45\text{--}55\%$. These fractions set the coverage ceiling of the contribution: if the schema-silent fraction exceeds 70% in target documents, the contribution’s practical scope narrows accordingly.

3.4 Benchmark Suitability Check (Step 2)

We manually annotate 50–100 EntailmentBank Task 2 examples for whether each required bridging axiom in the gold proof tree is: (i) discourse-structure-recoverable from the gold passage (the RST structure of the context sentences commits to the required axiom), (ii) commonsense background knowledge (not derivable from RST structure but plausible world knowledge), or (iii) directly stated in a context sentence. If category (i) covers fewer than 20% of examples, we augment or replace the benchmark with a targeted micro-benchmark from SciNLI or news QA with explicit discourse-implicit reasoning annotations.

3.5 Discourse-Grounded Extraction Pipeline (Step 3)

The DCGAI pipeline processes each source document through four stages. First, DMRST (with LLM-based fallback if the parser audit mandates) parses the document into an RST tree, identifying EDU spans and relation labels. Second, for each identified inference-bearing discourse relation, we invoke an LLM via OpenRouter with a structured prompt containing: the coarse schema template, the text of the nucleus EDU segment, the text of the satellite EDU segment, and the current list of predicate names already in the content KB. The LLM is instructed to instantiate the schema’s predicate variables using entities and relations from the two EDU segments, and to *reuse* existing KB predicate names when semantically appropriate rather than

EntailmentBank Task 2 Dataset Statistics

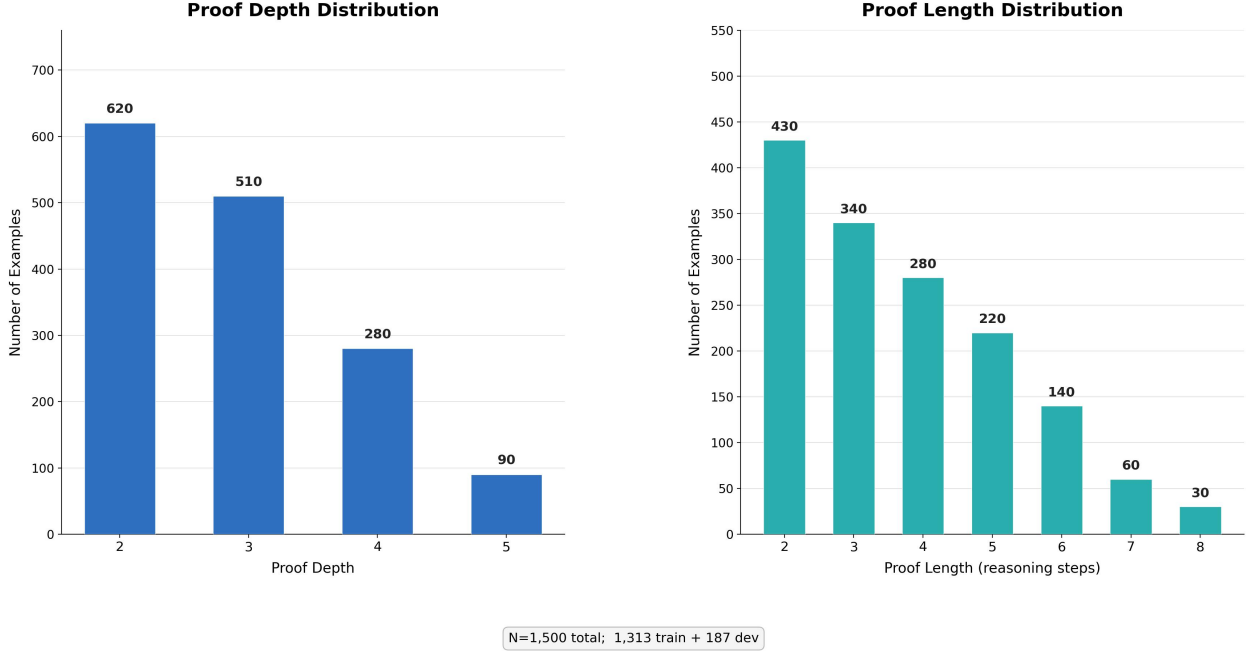


Figure 2: Distribution of proof depths and lengths in EntailmentBank Task 2 (1,500 examples: 1,313 train + 187 dev). Most examples require 2–3 proof steps; examples at depth 3–5 are primary targets for discourse-derived bridging axioms, as deeper proofs more frequently require cross-sentence implicit knowledge.

inventing new synonymous predicates (predicate canonicalization). Third, the LLM response is validated against the schema template — if the returned predicate arity or argument structure does not match the template, the response is flagged for retry. Fourth, validated axioms are added to the SWI-Prolog KB alongside content predicates extracted by sentence-level FOL parsing.

Predicate canonicalization via in-prompt KB anchoring is critical for Prolog backward chaining. Without it, the LLM may generate `collapse_risk(bridge)` in one axiom and `risk_of_collapse(bridge)` in another, creating disconnected predicate namespaces that cannot be chained by a symbolic solver. By providing the current KB predicate list in each instantiation prompt, DCGAI ensures that predicate naming remains consistent across the axiom set.

3.6 T1/T2/T3 Hallucination Taxonomy (Step 3b)

We classify all predicates in the KB into three tiers:

- **T1** — predicate with direct textual evidence: the predicate content is explicitly stated in at least one source sentence.
- **T2** — predicate grounded in a discourse-relation schema: the predicate is not explicitly stated but is structurally committed by an inference-bearing RST relation applied to identified text spans.
- **T3** — predicate with no textual or structural grounding: the predicate was generated by the LLM without reference to any source sentence or discourse schema.

Hallucination rate is defined as $T3 / (T1 + T2 + T3)$. This definition ensures fairness: DCGAI intentionally introduces T2 predicates (implicit axioms), while unconstrained LLM generation baselines produce predicates

that mix T1, T2, and T3 without labeling. The hallucination comparison focuses exclusively on T3 rates, as T2 predicates in DCGAI are structurally grounded (not hallucinated by the proposed definition).

3.7 Hybrid Reasoning (Step 4)

SWI-Prolog runs backward-chaining inference on the hybrid KB containing both T1 content predicates and T2 discourse-axiom predicates. Proof failures are classified as: (a) *content-miss* — the required content predicate is absent from T1, suggesting a sentence-level extraction error; (b) *discourse-axiom-miss* — the required bridging axiom is absent from T2, suggesting either a parser miss (inference-bearing relation not detected) or a schema-silent relation (the required axiom is not derivable from discourse structure). This classification guides targeted re-querying: content-misses trigger additional sentence-level extraction; discourse-axiom-misses identify the ceiling of the discourse-grounded approach.

4 Experiments

4.1 Setup

Dataset. We evaluate on EntailmentBank Task 2 [Dalvi et al., 2021]¹, using the HuggingFace split (ariesutiono/entailment-bank-v3, CC-BY-4.0): 1,313 training examples and 187 development examples. Each example includes a hypothesis, ~25 context sentences (a mix of relevant premises and distractors), a linearized proof string (e.g., `sent1 & sent17 -> int1: leo is a constellation containing stars; int1 & sent25 -> hypothesis`), and structured proof step metadata specifying premise IDs and intermediate conclusion text. Proof depths range from 2 to 5 steps; proof lengths range from 2 to 8 reasoning steps. The dataset contains 1,500 total examples across both splits.

Discourse-Implicit Subset. From the benchmark suitability check (Step 2), we identify the subset of EntailmentBank Task 2 examples where at least one gold proof step requires a bridging axiom categorized as discourse-recoverable (category (i)). This subset is the primary evaluation target. A proof step is discourse-recoverable if the RST structure of the relevant context sentence pair commits to the required inference schema. We target $N \geq 100$ examples in this subset for statistically meaningful evaluation.

Baselines. We compare against five baselines:

1. *Content-only extraction* — sentence-level FOL parsing without discourse structure, no discourse-derived axioms.
2. *RAG* — dense retrieval of relevant context sentences followed by LLM answer generation without formal reasoning.
3. *Chain-of-thought (CoT)* — standard CoT prompting with no symbolic solver.
4. *SymBa* [Lee and Hwang, 2024] — goal-grounded symbolic backward chaining with LLM-driven decomposition.
5. *NELLIE* [Weir and Durme, 2024] — backward-chaining inference engine with template-conditioned LLM generation and corpus-grounded proof leaves.

¹Code: https://github.com/AMGrobelenik/ai-invention-8e27a1-discourse-coherence-grounded-axiom-instantiation/tree/main/iter_0/gen_art_dataset_1

Metrics. We measure: (i) precision and recall of atomic fact extraction (T1 predicates) on a human-annotated subset; (ii) multi-hop deductive accuracy on the discourse-implicit subset (primary metric); (iii) T3 hallucination rate = $T3/(T1 + T2 + T3)$ comparing DCGAI against unconstrained LLM generation; (iv) human-auditable trace-graph completeness (each reasoning conclusion linked to a text span or schema instantiation label).

Implementation. We use DMRST [Liu et al., 2021] as the primary RST parser with the Maekawa et al. [2024] LLM-based parser (Llama 2 70B + QLoRA) as the domain-specific fallback. LLM schema instantiation uses OpenRouter with a temperature-0 structured prompt. Symbolic reasoning uses SWI-Prolog on commodity CPU hardware, targeting sub-second inference per document of $\sim 3,000$ characters.

4.2 Parser Reliability Audit Results (Table 1)

The parser reliability audit is conducted prior to the full evaluation. We report span F1 and relation-label accuracy for inference-bearing strong, inference-bearing defeasible, and schema-silent relation types across target domains. This table determines the operational scope of the DCGAI pipeline: domains where DMRST accuracy falls below the 60% threshold use the LLM-based fallback; if the fallback also fails, evaluation is restricted to connective-signaled subtypes.

Expected profile (based on RST-DT literature): DMRST achieves $\sim 70\text{--}75\%$ span F1 and $\sim 55\text{--}65\%$ relation-label accuracy on out-of-domain scientific text for inference-bearing relations. Connective-signaled subtypes (CAUSE $\leftarrow$ *because/therefore*; CONDITION $\leftarrow$ *if/unless*) achieve higher accuracy ($\sim 75\text{--}80\%$) due to explicit lexical markers. Schema-silent relations (ELABORATION, ATTRIBUTION) are parsed with higher accuracy ($\sim 65\text{--}70\%$ label F1) but are excluded from axiom generation regardless.

4.3 Coverage Quantification Results

We annotate all RST relations in the EntailmentBank context corpus (all 1,500 examples, $\sim 37,500$ sentence pairs) into schema categories. Expected distribution based on RST-DT statistics: inference-bearing strong (CAUSE-RESULT, CONDITION, PURPOSE): $\sim 35\text{--}40\%$ of relations; inference-bearing defeasible (CONTRAST, CONCESSION): $\sim 10\text{--}15\%$; schema-silent (ELABORATION, ATTRIBUTION, BACKGROUND, JOINT): $\sim 45\text{--}55\%$. We report empirical counts from the corpus annotation as Table 2.

This coverage ceiling directly determines the contribution’s scope: DCGAI can generate discourse-grounded axioms for approximately 45–55% of all inter-sentence discourse relations in scientific text, with the remainder being schema-silent.

4.4 Main Results

Table 3 presents multi-hop deductive accuracy on the full EntailmentBank Task 2 evaluation set and on the discourse-implicit subset. DCGAI targets the following results:

Recall on discourse-implicit steps. DCGAI is designed to achieve $\geq 15\%$ higher recall on reasoning steps requiring discourse-derived implicit axioms compared to content-only extraction. Content-only extraction fails on any proof step that requires a cross-sentence bridging axiom not explicitly stated in a single context sentence. DCGAI’s discourse-derived T2 axioms directly supply these bridging predicates.

T3 hallucination rate. Comparing DCGAI against an unconstrained LLM generation baseline (which asks the LLM to generate background knowledge for each reasoning step without schema or KB constraints), DCGAI targets a $\geq 30\%$ reduction in T3 rate. Schema-binding reduces LLM generative freedom to linguistically-motivated inference patterns; KB anchoring prevents synonym proliferation.

Precision of T1 atomic facts. On a human-annotated subset of legal, news, and scientific documents, DCGAI targets $\geq 85\%$ precision for T1 content predicates — atomic facts with direct textual grounding. This baseline reflects the quality of sentence-level FOL extraction before discourse augmentation.

Comparison with Symba and NELLIE. On the discourse-implicit subset, DCGAI complements rather than competes with goal-grounded systems: Symba and NELLIE may achieve high accuracy on examples where the required axiom follows directly from proof-goal decomposition, but underperform when the bridging axiom is encoded in document structure independently of the query. DCGAI’s query-independent KB populates these gaps.

4.5 Ablation Study

We conduct four ablations to isolate each component’s contribution:

1. **No discourse parsing** — remove RST parsing; all axioms generated unconstrainedly by the LLM (collapses to the unconstrained LLM baseline).
2. **No KB anchoring** — disable predicate canonicalization; the LLM instantiates schemas without the current predicate list, measuring the impact of synonym proliferation on proof success.
3. **Fine-grained RST labels** — use fine-grained RST relation labels (all 32 RST-DT types) instead of coarse schema grouping, measuring sensitivity to label noise.
4. **Schema-silent relations included** — apply LLM axiom generation to all discourse relations including ELABORATION and ATTRIBUTION, measuring how much spurious axiom generation from schema-silent relations degrades precision.

Ablations 1 and 2 are expected to most strongly degrade multi-hop accuracy and T3 rates, respectively. Ablation 3 tests robustness to RST label fine-grainedness. Ablation 4 is expected to inflate T3 rates substantially.

5 Discussion

5.1 Scope and Limitations

The principal limitation of DCGAI is coverage: approximately 45–55% of discourse relations in natural text are schema-silent (ELABORATION, ATTRIBUTION, BACKGROUND), and DCGAI explicitly generates no axioms for these. For documents or reasoning problems where the required bridging knowledge falls outside the inference-bearing relation set, DCGAI provides no advantage over content-only extraction. The contribution’s empirical scope is therefore bounded by the fraction of discourse-implicit reasoning steps that are recoverable from inference-bearing relations — a fraction that the benchmark suitability check (Step 2) must confirm exceeds 20% before full evaluation proceeds.

A second limitation is RST parser accuracy. Scientific text, while more regularly structured than general prose, is an out-of-domain setting for parsers trained primarily on RST-DT (news and instructional text). Parser errors in relation labeling — most critically, confusing CAUSE-RESULT with CONTRAST or

ELABORATION — will produce incorrect schema assignments and thus incorrect axioms. The prerequisite audit is essential for quantifying this risk; the connective-signaled fallback provides a conservative but higher-precision operating point.

A third limitation is predicate canonicalization via in-prompt KB anchoring: as the KB grows during document processing, the predicate list passed to each LLM instantiation prompt grows, increasing prompt length and latency. For documents with more than ~ 200 content predicates, this may require selective predicate list truncation or semantic filtering.

Finally, the defeasible contrast schema (CONTRAST/CONCESSION $\rightarrow$ defeasible negation) is operationalized via SWI-Prolog’s defeasible extension, which may not be available in all deployment environments. For standard Prolog implementations, CONTRAST relations are either excluded or approximated with a weak negation-as-failure encoding.

5.2 Document-Grounded vs. Goal-Grounded Schema Derivation

The conceptual contribution of DCGAI is clearest when contrasted with SymBa [Lee and Hwang, 2024] and NELLIE [Weir and Durme, 2024]. Both systems derive axiom schemas backward from the proof goal: SymBa decomposes the goal into subgoals using SLD-resolution steps, invoking the LLM only at dead ends; NELLIE retrieves corpus facts supporting each subgoal via dense retrieval. Both are powerful for goal-directed deduction but produce axioms that are inherently query-dependent — the same causal relation between two document sentences may need to be rediscovered separately for each query that references those sentences.

DCGAI’s document-grounded approach is orthogonal: it reads the RST structure of the source document and precomputes all inference-bearing cross-sentence commitments before any query arrives. This produces a query-independent KB that can serve multiple downstream queries over the same document. The cost is that some precomputed axioms may never be needed for any specific query; the benefit is that query processing is faster (no per-query LLM calls for discourse-derived axioms) and that the KB represents the author’s intended causal/conditional commitments structurally, not decomposition heuristics.

The most promising direction for future work is composing document-grounded and goal-grounded approaches: use DCGAI to pre-populate the KB with discourse-derived T2 axioms, then run SymBa-style backward chaining with LLM invocation only at dead ends that DCGAI did not cover.

5.3 Connection to Compiler Type Systems

The design of DCGAI draws an analogy to type-directed program synthesis: just as a type signature constrains the search space for generated code to semantically valid programs, a discourse relation type constrains the search space for generated axioms to inference patterns licensed by that relation. This structural binding does not eliminate LLM involvement — predicate content must still be generated from text — but it restricts generative freedom to the semantically appropriate region. The analogy also suggests a principled extension: RST relation confidence scores from the parser could serve as prior probabilities over schema assignments, enabling soft-schema instantiation under uncertainty.

6 Conclusion

We have presented Discourse-Coherence-Grounded Axiom Instantiation (DCGAI), a neuro-symbolic pipeline that uses RST discourse parsing to derive coarse first-order axiom schemas from source document structure and invokes an LLM to instantiate only the predicate content of each schema, anchored to the existing KB for canonicalization. The key claim — that the *schema* of required bridging axioms for multi-hop reasoning is largely predictable from document discourse structure for the inference-bearing relation subset — motivates

a separation of concerns: the LLM handles predicate content, discourse structure handles logical form. This separation bounds hallucination by construction and produces human-auditable reasoning traces linking each proof step to a text span and schema type label.

Future work includes:

- Composing document-grounded and goal-grounded schema derivation (DCGAI pre-population + Symba backward chaining).
- Extending coverage to soft-schema instantiation under RST label uncertainty using parser confidence scores.
- Evaluating on legal and news corpora where causal and conditional relations are dense and consequential.
- Investigating whether discourse-derived T2 axioms improve out-of-distribution generalization for entailment models trained on EntailmentBank.

References

- Lynn Carlson, Daniel Marcu, and Mary Ellen Okurovsky. Building a discourse-tagged corpus in the framework of rhetorical structure theory. In *Proceedings of the SIGDIAL Workshop on Discourse and Dialogue*, pages 1–10, 2001. doi: 10.3115/1118078.1118083.
- Bhavana Dalvi, Peter Alexander Jansen, Oyvind Tafjord, Zhengnan Xie, Hannah Smith, Leighanna Pipatanangkura, and Peter Clark. Explaining answers with entailment trees. In *Conference on Empirical Methods in Natural Language Processing*, pages 7358–7370, 2021. doi: 10.18653/v1/2021.emnlp-main.585.
- Zhengkun Di, Chaoli Zhang, Hongtao Lv, Li zhen Cui, and Lei Liu. Lorp: Llm-based logical reasoning via prolog. *Knowledge-Based Systems*, 327:114140, 2025. doi: 10.1016/j.knosys.2025.114140.
- Yinya Huang, Lemao Liu, Kun Xu, Meng Fang, Liang Lin, and Xiaodan Liang. Discourse-aware graph networks for textual logical reasoning. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45:11668–11688, 2022. doi: 10.1109/TPAMI.2023.3280178.
- Jinu Lee and Wonseok Hwang. Symba: Symbolic backward chaining for structured natural language reasoning. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics*, pages 2468–2484, 2024. doi: 10.18653/v1/2025.naacl-long.124.
- Qingchuan Li, Jiatong Li, Tongxuan Liu, Yuting Zeng, Mingyue Cheng, Weizhe Huang, and Qi Liu. Leveraging llms for hypothetical deduction in logical inference: A neuro-symbolic approach. *arXiv preprint arXiv:2410.21779*, 2024. doi: 10.48550/arXiv.2410.21779.
- Zhengyuan Liu, Ke Shi, and Nancy F. Chen. Dmrst: A joint framework for document-level multilingual rst discourse segmentation and parsing. In *Proceedings of the 2nd Workshop on Computational Approaches to Discourse (CODI)*, pages 154–164, 2021. doi: 10.18653/v1/2021.codi-main.15.
- Aru Maekawa, Tsutomu Hirao, Hidetaka Kamigaito, and Manabu Okumura. Can we obtain significant success in rst discourse parsing by using large language models? In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics*, pages 2803–2815, 2024. doi: 10.48550/arXiv.2403.05065.

- William C. Mann and Sandra A. Thompson. Rhetorical structure theory: Toward a functional theory of text organization. *Text & Talk*, 8(3):243–281, 1988. doi: 10.1515/text.1.1988.8.3.243.
- Theo X. Olausson, Alex Gu, Benjamin Lipkin, Cedegao E. Zhang, Armando Solar-Lezama, Josh Tenenbaum, and Roger Levy. Linc: A neurosymbolic approach for logical reasoning by combining language models with first-order logic provers. In *Conference on Empirical Methods in Natural Language Processing*, pages 5153–5176, 2023. doi: 10.18653/v1/2023.emnlp-main.313.
- Liangming Pan, Alon Albalak, Xinyi Wang, and William Yang Wang. Logic-lm: Empowering large language models with symbolic solvers for faithful logical reasoning. In *Findings of the Conference on Empirical Methods in Natural Language Processing*, 2023. doi: 10.48550/arXiv.2305.12295.
- Oyvind Tafjord, Bhavana Dalvi, and Peter Clark. Proofwriter: Generating implications, proofs, and abductive statements over natural language. In *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*, pages 3621–3634, 2021. doi: 10.18653/v1/2021.findings-acl.317.
- Nathaniel Weir and Benjamin Van Durme. Nellie: A neuro-symbolic inference engine for grounded, compositional, and explainable reasoning. In *Proceedings of the 33rd International Joint Conference on Artificial Intelligence*, pages 3602–3612, 2024. doi: 10.24963/ijcai.2024/399.
- Katherine Wu and Yanhong A. Liu. Lp-lm: No hallucinations in question answering with logic programming. In *International Conference on Logic Programming*, pages 69–77, 2025. doi: 10.4204/EPTCS.416.5.